

Food Delivery Analytics — Data Visualization

Loading the cleaned dataset and creating charts to visualize the key findings.

Step 1: Load cleaned data

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('../data/food_delivery_cleaned.csv')
df['OrderTimestamp'] = pd.to_datetime(df['OrderTimestamp'])

sns.set_style('whitegrid')

print(df.shape)
df.head()
```

(49990, 24)

Out[1]:

	OrderID	CustomerID	RestaurantID	OrderTimestamp	OrderAmount	DiscountAmount
0	O000001	C03175	R0115	2025-09-28	1079.36	107.94
1	O000002	C02432	R0077	2024-02-13	1499.63	149.96
2	O000003	C01133	R0131	2024-03-07	597.18	59.72
3	O000004	C04813	R0109	2025-01-27	1257.48	125.75
4	O000005	C04948	R0001	2024-10-25	1010.57	0.00

5 rows × 24 columns



Chart 1: Order Status Breakdown

Shows the ~40% order failure rate — the project's core finding.

```
In [3]: status_counts = df['OrderStatus'].value_counts()

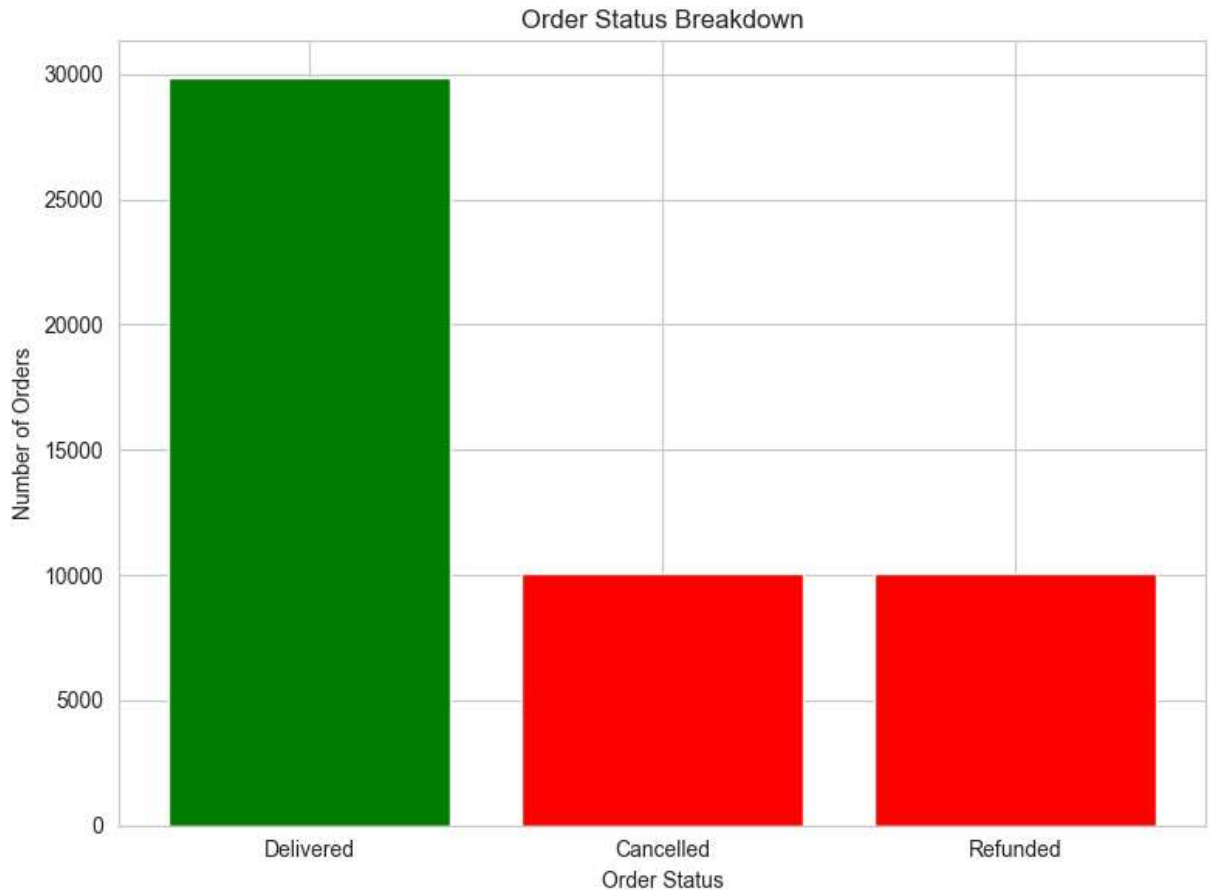
colors = []
for status in status_counts.index:
    if status == 'Delivered':
        colors.append('green')
    else:
```

```

colors.append('red')

plt.figure(figsize=(8, 6))
plt.bar(status_counts.index, status_counts.values, color=colors)
plt.title('Order Status Breakdown')
plt.xlabel('Order Status')
plt.ylabel('Number of Orders')
plt.tight_layout()
plt.savefig('../Dashboard/chart1_order_status.png')
plt.show()

```



Finding: Nearly 40% of all orders fail to complete — 20.19% Cancelled and 20.14% Refunded, with only 59.67% successfully Delivered. This is the most critical operational issue in the dataset.

Chart 2: Revenue by Acquisition Channel

Shows that all channels perform similarly — no clear winner.

```

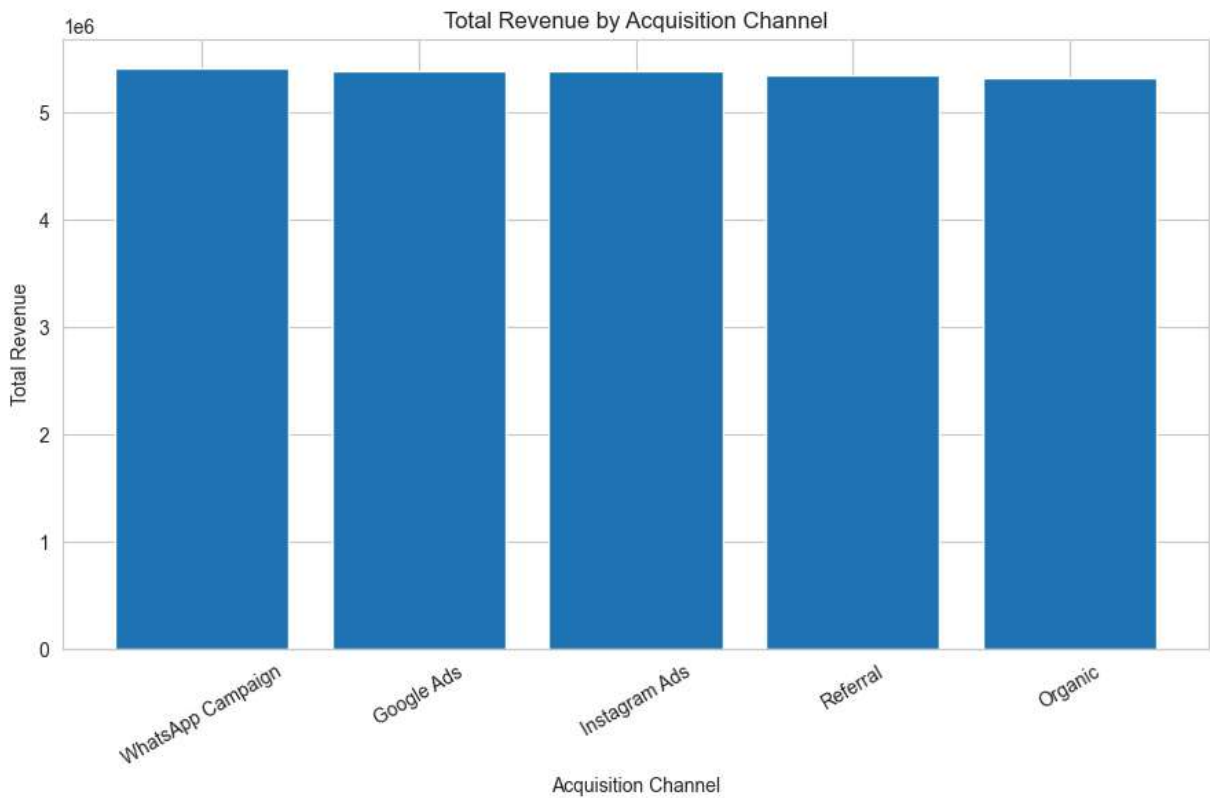
In [4]: delivered = df[df['OrderStatus'] == 'Delivered']

channel_revenue = delivered.groupby('AcquisitionChannel')['OrderAmount'].sum().sort

plt.figure(figsize=(9, 6))
plt.bar(channel_revenue.index, channel_revenue.values)
plt.title('Total Revenue by Acquisition Channel')

```

```
plt.xlabel('Acquisition Channel')
plt.ylabel('Total Revenue')
plt.xticks(rotation=30)
plt.tight_layout()
plt.savefig('../dashboard/chart2_channel_revenue.png')
plt.show()
```



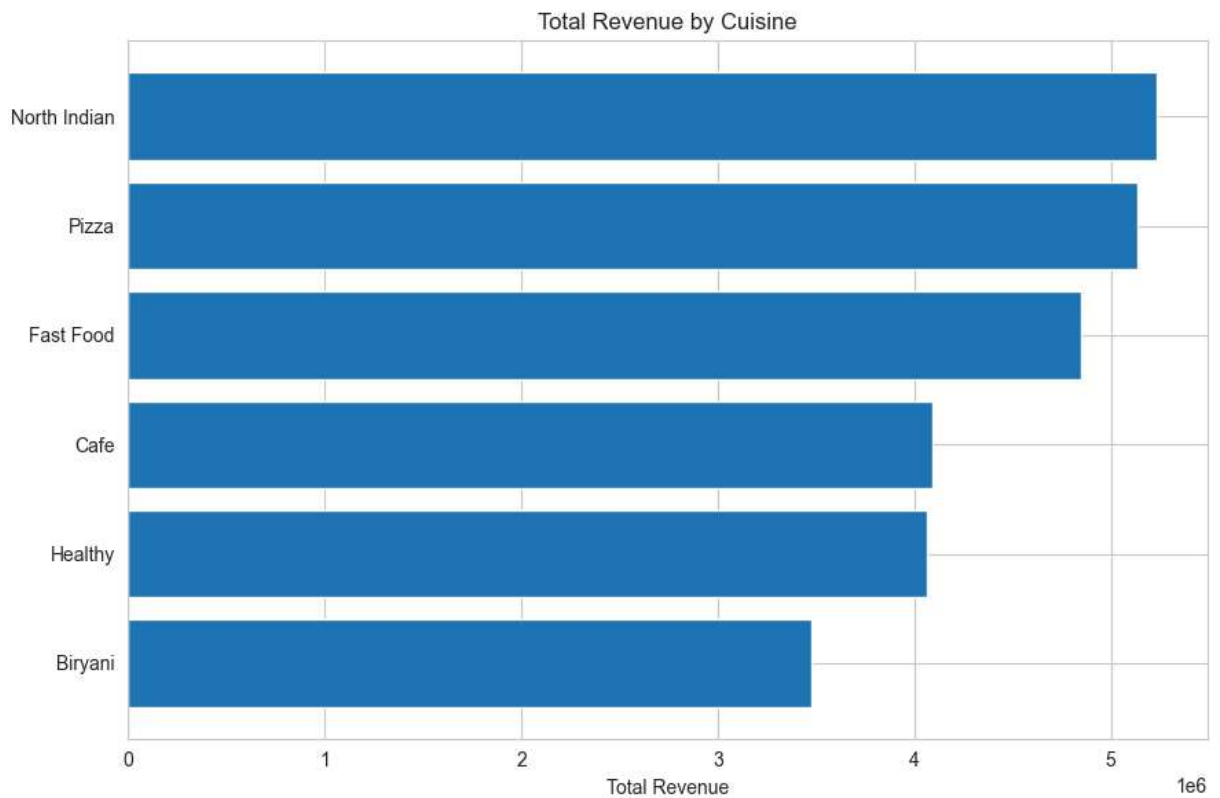
Finding: All 5 acquisition channels generate similar revenue (₹53-54L range), with no clear best-performing channel. Marketing spend allocation should be based on cost-per-acquisition rather than assumed revenue differences between channels.

Chart 3: Revenue by Cuisine

North Indian leads, but the gap between cuisines is modest.

```
In [5]: cuisine_revenue = delivered.groupby('Cuisine')['OrderAmount'].sum().sort_values(asc

plt.figure(figsize=(9, 6))
plt.barh(cuisine_revenue.index[::-1], cuisine_revenue.values[::-1])
plt.title('Total Revenue by Cuisine')
plt.xlabel('Total Revenue')
plt.tight_layout()
plt.savefig('../dashboard/chart3_cuisine_revenue.png')
plt.show()
```



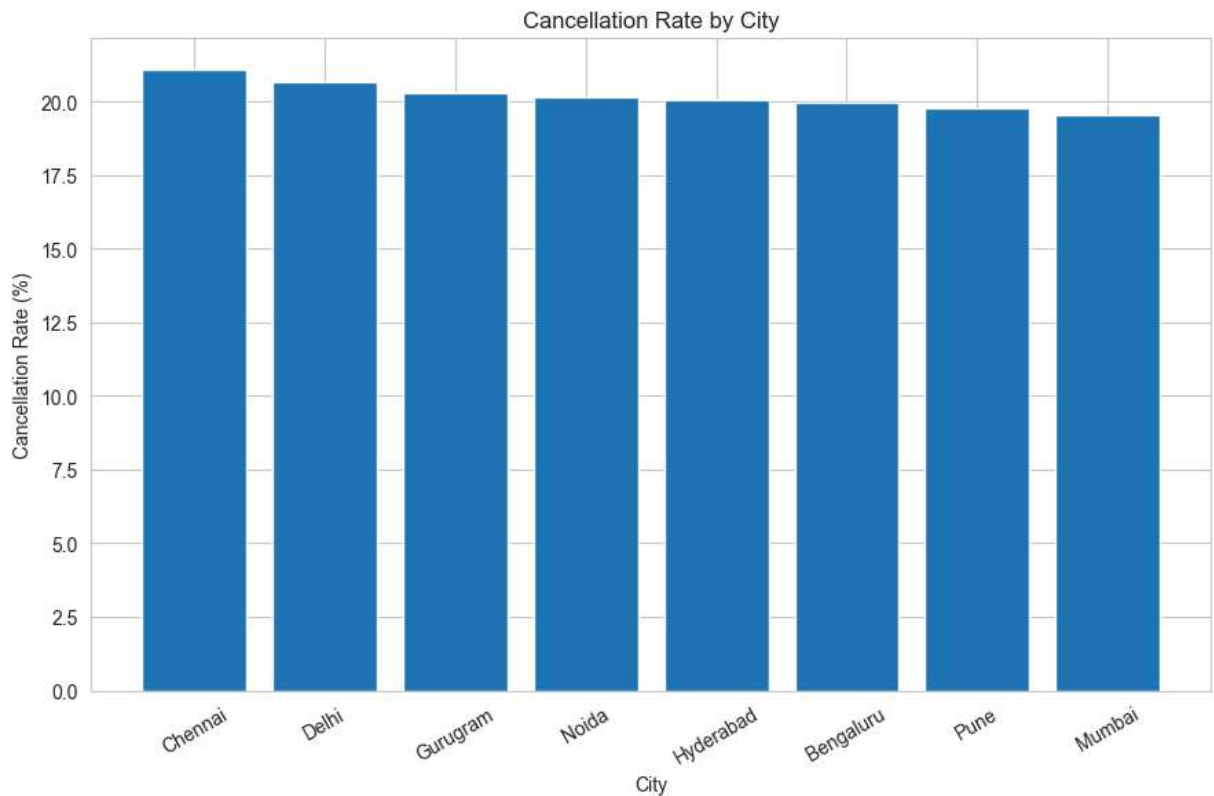
Finding: North Indian is the top-revenue cuisine, followed closely by Pizza and Fast Food. Revenue differences across cuisines are modest, and ratings don't strongly correlate with revenue.

Chart 4: Cancellation Rate by City

Shows Chennai has the highest cancellation rate, Mumbai the lowest.

```
In [6]: city_cancel = df.groupby('CustomerCity').agg(
        TotalOrders=('OrderID', 'count'),
        CancelledOrders=('IsCancelled', 'sum')
    )
    city_cancel['CancellationRate'] = city_cancel['CancelledOrders'] / city_cancel['TotalOrders']
    city_cancel = city_cancel.sort_values('CancellationRate', ascending=False)

    plt.figure(figsize=(9, 6))
    plt.bar(city_cancel.index, city_cancel['CancellationRate'])
    plt.title('Cancellation Rate by City')
    plt.xlabel('City')
    plt.ylabel('Cancellation Rate (%)')
    plt.xticks(rotation=30)
    plt.tight_layout()
    plt.savefig('../dashboard/chart4_city_cancellation.png')
    plt.show()
```



Finding: Chennai has the highest cancellation rate (~21%) while Mumbai has the lowest (~19.6%) — a modest but real gap worth investigating for city-specific delivery or restaurant capacity issues.

Chart 5: Monthly Revenue Trend

Shows how revenue moves over time across the dataset's date range.

```
In [7]: delivered['YearMonth'] = delivered['OrderTimestamp'].dt.to_period('M').astype(str)

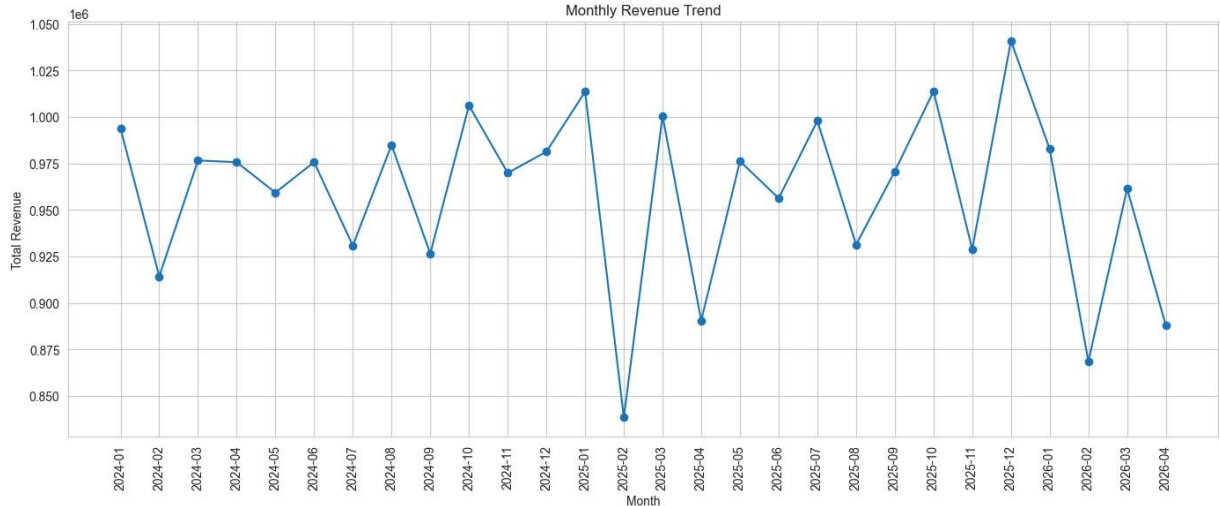
monthly_revenue = delivered.groupby('YearMonth')['OrderAmount'].sum()

plt.figure(figsize=(14, 6))
plt.plot(monthly_revenue.index, monthly_revenue.values, marker='o')
plt.title('Monthly Revenue Trend')
plt.xlabel('Month')
plt.ylabel('Total Revenue')
plt.xticks(rotation=90)
plt.tight_layout()
plt.savefig('../dashboard/chart5_monthly_revenue.png')
plt.show()
```

```
C:\Users\Mohit shrivastav\AppData\Local\Temp\ipykernel_20496\2649045047.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
delivered['YearMonth'] = delivered['OrderTimestamp'].dt.to_period('M').astype(str)
```



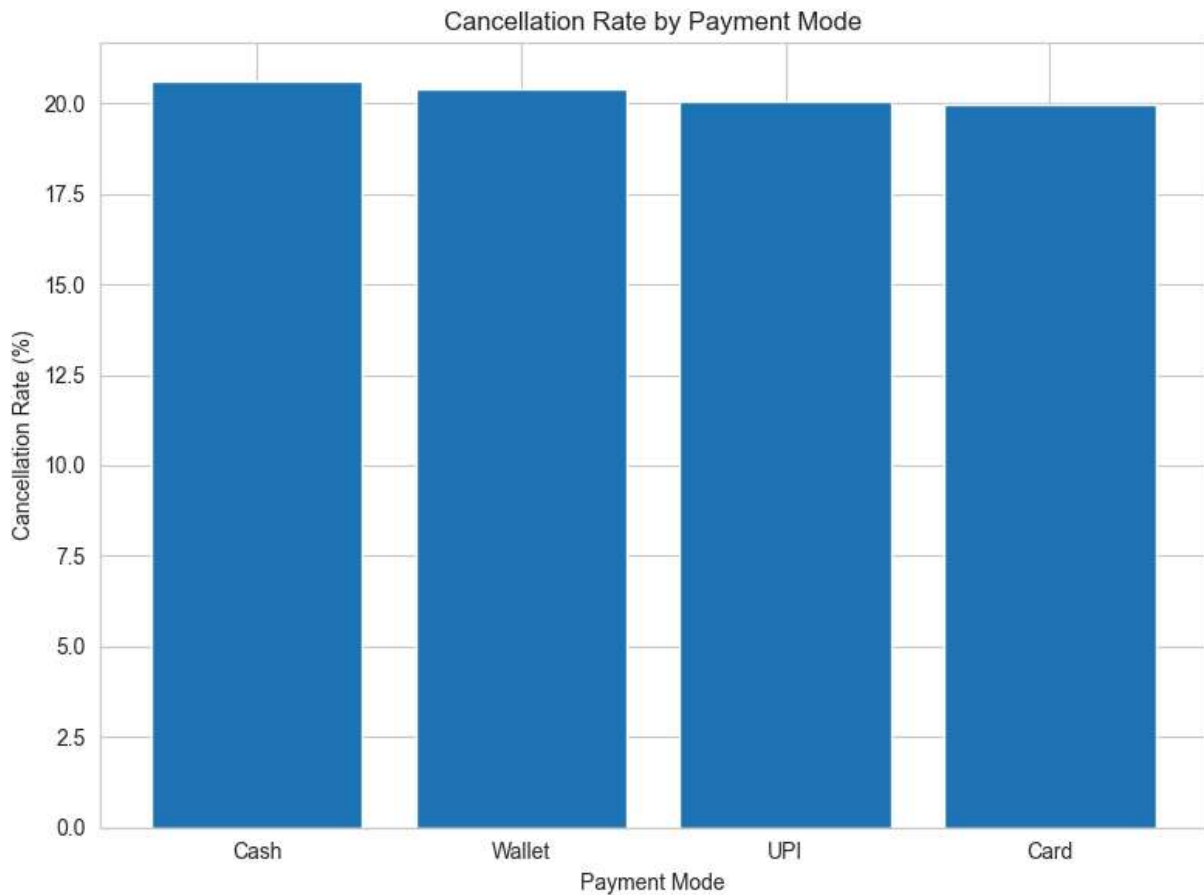
Finding: Monthly revenue fluctuates between roughly ₹840K and ₹1.04M with no clear long-term growth or decline trend. Unlike a retail dataset with obvious holiday seasonality, this data shows no consistent repeating pattern across months — revenue appears relatively stable overall, with some individual low months (e.g., February 2025, February 2026) that may be worth investigating further, though the ~2-year window is too short to confirm this as a true seasonal pattern rather than noise.

Chart 6: Cancellation Rate by Payment Mode

Shows cancellation rate is nearly flat across all payment methods.

```
In [9]: payment_cancel = df.groupby('PaymentMode').agg(
        TotalOrders=('OrderID', 'count'),
        CancelledOrders=('IsCancelled', 'sum')
    )
payment_cancel['CancellationRate'] = payment_cancel['CancelledOrders'] / payment_cancel['TotalOrders']
payment_cancel = payment_cancel.sort_values('CancellationRate', ascending=False)

plt.figure(figsize=(8, 6))
plt.bar(payment_cancel.index, payment_cancel['CancellationRate'])
plt.title('Cancellation Rate by Payment Mode')
plt.xlabel('Payment Mode')
plt.ylabel('Cancellation Rate (%)')
plt.tight_layout()
plt.savefig('../dashboard/chart6_payment_cancellation.png')
plt.show()
```



Finding: Cancellation rate is nearly identical across all payment modes (19.96%-20.63%), reinforcing that the ~40% order failure rate is a systemic issue rather than tied to a specific payment method.

Summary

This notebook validated the SQL analysis findings using Python and produced 6 visualizations covering order status, acquisition channel revenue, cuisine performance, city-level cancellation rates, monthly revenue trends, and payment mode cancellation consistency. Additional Python-only analysis revealed that cancellation rate is consistent across payment modes and cuisines, pointing to a systemic rather than segment-specific cause behind the high order failure rate.

Next step: Build an interactive Power BI dashboard connected to the SQL Server database.